MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS) REGULAR
EXAMINATION NOVEMBER 2024

SCHEME FOR EVALUATION

Programme Name	**B.Sc Forensic Science**
Course Title	GENERAL FORENSIC SCIENCE
Course Code	MG1DSCFSC100
QP Code	24900111

SECTION-A
Short Answer Type Questions
(Answer any 10 Questions out of 12)

1.	In sexual assault cases, serology helps identify and analyze bodily fluids like blood, semen, and saliva found at the crime scene or on the victim's clothing. This analysis can confirm the presence of these fluids. It aids in linking the suspect to the crime or supporting the victim's account. Modern DNA analysis often complements serology, providing stronger evidence for identification.
2.	Providing scientific analysis report (Forensic Report), expert testimony and objective opinions on evidence related to a case. They present their findings in a clear and unbiased manner, helping the court understand complex technical details. Their testimony can establish key facts, link suspects to crimes, or rule out false leads.
3.	Role of Director includes managing administrative and technical activities, supervising staff, and ensuring that high-quality forensic services are delivered efficiently. They establish policies, maintain accreditation standards, and oversee the proper handling, analysis, and reporting of evidence. The Director also liaises with law enforcement agencies, judiciary systems, and government authorities to ensure the lab supports criminal investigations and legal proceedings effectively.
4.	Role of DFSS include oversee and coordinate forensic science activities across various forensic laboratories in a country. DFSS ensures that forensic science laboratories maintain high standards of quality and efficiency in conducting forensic analysis. It sets guidelines, provides technical expertise, and offers training to forensic professionals.
5.	Crime Detection, Evidence Examination, Justice Support, Crime Prevention, Identification, Reconstruction, Standardization, report writing (Including brief explanation of any of the 3 keywords above or similar)
6.	The law of individuality is a fundamental principle in forensic science that states that every object, whether natural or man-made, has unique identifying features that cannot be duplicated. This means that no two objects can be exactly the same, even if they appear similar.
7.	Forensic toxicology is the scientific study of toxins, drugs, and

	poisons, focusing on their detection, effects, and role in legal cases. It involves analyzing biological samples such as blood, urine, and tissues to determine the presence of harmful substances and their impact on the body**.**
8.	A crime scene is any location or object that may contain evidence of a crime, including the place where the crime actually took place.
9.	Expert evidence is information or opinion given by an expert in any field that person is specialized in, which comes out to be evidence in any matter. In field of law, expert witness is a person whose opinion is accepted by judge relating to any fact or evidence.
10.	Functions of Prisons and Correctional Institutions -The functions of prisons shall be as follows: (i) to keep in safe custody, a prisoner committed to it under any writ, warrant or by the order of any court or any other competent authority; (ii) to take suitable measures for the safety and security of prisoners; (iii) to provide the prisoners with food, clothing, accommodation, other necessities and medical treatment; (iv) to provide correctional treatment to the prisoners with the objective of rehabilitating them in the society as law abiding citizens; (v) to maintain discipline in the prison in accordance with the provisions of this Act and the rules made thereunder. (Any 2 with similar meaning)
11.	ISO/IEC 17025 is the international standard for testing and calibration laboratories. It sets out requirements for the competence, impartiality, and consistent operation of laboratories, ensuring the accuracy and reliability of their testing and calibration results. This standard is vital for laboratories as it enhances the credibility of their testing and calibration work, fostering trust among clients and regulatory authorities. Compliance with ISO/IEC 17025 demonstrates a laboratory's commitment to quality, technical proficiency, and scientific rigor.
12.	Judicial review The power to declare laws or ordinances unconstitutional if they violate the Indian Constitution. Original jurisdiction The power to hear disputes in matters that are not usually appealed, such as: Marriage, divorce, and will, Company laws, Election of MLAs and MPs , Admiralty, Contempt of court. Writ jurisdiction The power to issue writs to enforce fundamental rights and other legal rights. Appellate jurisdiction The power to review judgments from lower courts, including the power to reduce sentences or acquit the accused. (Any 2)

Part B
Long Answer Type Questions
(Answer any 5 Questions out of 7)

13.	Ethics in forensic science is critical to maintaining the integrity of the justice system. Forensic scientists often deal with evidence that can significantly influence legal outcomes, and their responsibilities require strict adherence to ethical principles to ensure fairness, accuracy, and impartiality. 1. Principle of Impartiality, 2. Accuracy and Precision in Analysis, 3. Honesty and Integrity, 4. Confidentiality, 5. Competency and Continuing Education, 6. Avoiding Misleading or Incomplete Testimony, 7. The Principle of Professionalism. (explained)
14.	• Title page (Case No., PS, IO details, nature of crime etc) • Table of contents • Reason for the investigation • Signature page • Evidence examined (answers to the questions submitted by IO) • Description of the investigation • Details of the findings (methods and findings) • Conclusions
15.	Forensic science plays a crucial role in criminal investigations by providing scientific methods and evidence that can establish facts, identify suspects, and assist in the judicial process. It bridges the gap between law enforcement and science, offering objective, reliable, and precise information crucial for solving crimes. Identification of Suspects, Reconstruction of Crime Scenes, Authentication of Evidence, Establishing Time of Death etc. (Any relevant example may be considered)
16.	Forensic Photography involves capturing accurate images of crime scenes, evidence, and accidents to aid in criminal investigations. It ensures that visual documentation of physical evidence is preserved for later analysis. **Application:** Documenting crime scenes for investigation and court presentations. Capturing images of injuries, weapon marks, and other physical evidence. Assisting in crime scene reconstruction through visual data.
17.	Preservation of evidence is vital in forensic investigations as it ensures that the evidence remains intact, unaltered, and reliable throughout the investigative and judicial processes. Proper handling and preservation maintain the evidentiary chain of custody, safeguard the evidence from contamination, and uphold its admissibility in court.
18.	Forensic Science Laboratories (FSLs) are specialized institutions that assist in criminal investigations by analyzing physical, chemical, and biological evidence. Their organizational structure is designed to ensure efficient processing, analysis, and reporting of forensic evidence while maintaining high standards of scientific accuracy and impartiality. Director-Additional Director-Joint Director (Sections)-Senior Scientific Officer-Scientific Officer- Junior Scientific Officer-Lab assistant (explained in detail)

19	Mobile Forensic Vehicles are specially customized vehicles used br crime scene investigators to process the crime scene which provide exceptional functional capabilities to investigators providing facilities such as extended space, lightening, build cabinetry, mobile forensic/crime lab with workstation for identification, examination, collection storage of evidence right at the crime scene. In this article, we will look into the importance of the Mobile forensic Vehicle, its use and benefits in crime scene investigation management. Benefits-Reduced Response Time, On Scene Spot Test/ Field Test, Evidence identification and Collection, Evidence Storage and Transport.
Part C Essay Type Questions (Answer any 1 Question out of 2)	
20	Forensic Chemistry When cases that involve chemicals and chemical reactions are investigated, forensic chemistry is used. It deals with all kinds of adulterations and examines the evidence to make inferences regarding the case. The adulterations that it deals with are food, liquor, petroleum, explosive materials, alcoholic and non-alcoholic beverages, carbonated drinks, packaged juices, and drugs. Apart from these it also examines the adulteration of chemicals and compounds like RDX, PETN, HMX, Nitrogylcerine, Nitrocellulose, Sulphur Oxides, Fuses, Cordite sticks, etc. Forensic Biology Forensic Biology deals with biological evidence-like tissues, blood, muscles, semen, hair, saliva, pollen, urine, etc. It is one of the most important branches of forensics as it deals with living beings. Digital Forensics Digital Forensics involves the recovery, analysis, and investigation of data found in digital devices such as computers, smartphones, and other electronic systems, often for use in criminal investigations. • Recovering deleted files, emails, and chat logs from digital devices. • Analyzing cybercrimes, such as hacking, fraud, or identity theft. • Presenting digital evidence in court to support criminal investigations. (Appropriate and relevant case studies may be considered)
21	The imprisonment method which originally means detention till the trial is over or a debt was paid is now frequently used for all crimes except light misdemeanors. With a change in time, prison functions have also improved from supervisory, to forcible, to corrective. Earlier prisons were mainly used as penal institutions. No consideration was given to differences in age, gender, type of offence, criminal history or psychological state of the criminals. After the references of the Jail Reforms Committee of 1919 the situation in prisons has progressively improved and they have come to be used for corrective purposes, i.e. via teaching offenders the moral and social values through education, work discipline and

	moral instructions which help them shaping up as normal citizens and law-abiding citizens. The elementary purposes of jails nowadays in India are: **1** ***Social isolation and confinement:*** to separate a criminal from the normal society‟ (or law abiding citizens) because he has proved to be a threat to its organization, stability, and cohesion, and to keep him out of circulation and so securely confined that his deviation from law does not disturb the harmony of society. **2** ***Repentance:*** to keep an offender in an isolated place where he could ponder over the consequences of his wrong deeds. **3** ***Punishment and deterrence:*** to cause some pain and suffering, on an offender (i.e. by way of primitive actions) for violating legal norms, so that criminals are worse off than the poorest of honest citizens; law-abiding individuals are satisfied that law-breakers are penalized and they are being protected against the threat of recidivism; and members of society may be deterred from committing crimes. The society's moral fabric is restored. **4** ***Protection:*** guarding public from prisoners by pointing out persons who violate laws and stigmatizing them so that others are warned against them. **5** ***Reformation:*** to change offender's values, motivations, attitudes and perceptions and to resocialise him according to the norms of the community. According to Tappan, (1951), "The objective of prison education in its largest sense would be the socialization of the convicts by diverse vulnerable and expressional doings with stress on person inmate's needs. The aim of this programme is to return of these convicts to society with an aspiration to work themselves as good citizens and with the skill and knowledge that will gave them an equitable opportunity to uphold themselves and their children by honest labor". Details of -Prison -Open Jails – Juvenile homes etc may be considered